

REQUEST FOR PROPOSAL & STATEMENT OF WORK

Customer Churn Prediction Platform — TelecomPlus Pvt Ltd

Client: TelecomPlus Pvt Ltd | **Vendor:** DataMinds Analytics | **Budget:** \$420,000 | **Duration:** 8 months | **Start:** 2026-09-01

1. Project Overview & Goals

TelecomPlus requires a real-time customer churn prediction platform processing 8 million subscriber records monthly, integrating with CRM and billing systems, and generating automated retention campaigns via a REST API. The model must be explainable (SHAP values) for regulatory purposes.

Goals: (1) Model AUC-ROC ≥ 0.87 on held-out test set. (2) Prediction latency $<200\text{ms}$ per customer. (3) Reduce churn rate by 15% within 6 months of deployment. (4) Model bias audit passed (no demographic discrimination).

2. Scope of Work & MVP

MVP (Month 1–3): Batch churn prediction pipeline, basic feature engineering (usage, billing, tenure), and CSV export for CRM upload. MVP complete when model AUC ≥ 0.82 on validation set and batch job runs without failure for 2 weeks.

Out of Scope: Real-time streaming beyond defined API; NLP on call recordings; multi-language support for explainability reports.

3. Work Breakdown & Timelines

ID	Work Item	Timeline	Entry Criteria	Definition of Done / Exit Criteria
WI-001	Data Pipeline & Feature Store	Month 1–2	Data access agreements signed; schema finalized	ETL pipeline processes 8M records $<4\text{h}$; feature store versioned
WI-002	Churn Prediction Model (Batch)	Month 2–4	WI-001 done; feature set approved	AUC-ROC ≥ 0.87 ; Precision ≥ 0.80 at 30% recall threshold; SHAP values available
WI-003	Real-Time Prediction API	Month 3–5	WI-002 done; API contract approved	REST API $<200\text{ms}$ p99; autoscales to 500 RPS; versioned endpoints
WI-004	CRM & Campaign Integration	Month 5–7	WI-003 done; CRM API credentials provided	Model predictions pushed to CRM daily; campaign trigger fires within 1h
WI-005	Monitoring, Drift, Retraining	Month 6–8	WI-002 deployed; monitoring tooling installed	Dashboard shows metrics; automated drift alerts; automated retraining when AUC drops $>2\%$

4. Legal & Contractual Obligations

- 4.1 Late Delivery Penalty:** 1% of milestone value per calendar week of delay. Maximum 12% of total contract value. Penalty waived if delay is caused by Client's failure to provide data access within agreed SLA.
- 4.2 Subcontracting:** No subcontracting permitted. All model development, data engineering, and deployment must be performed by Vendor's permanent employees listed in Schedule A. Violation is material breach.
- 4.3 Indemnity:** Client owns all subscriber data and indemnifies Vendor against claims arising from the data itself. Vendor indemnifies Client against algorithmic discrimination findings and model-related regulatory penalties.
- 4.4 Intellectual Property:** All trained models, feature engineering pipelines, and associated code are works-for-hire. Client owns all deliverables. Vendor may not use TelecomPlus subscriber data or derived features for any other client. Vendor retains ownership of generic ML framework code.
- 4.5 Insurance:** Professional Indemnity $\geq \$2\text{M}$; Cyber Liability $\geq \$3\text{M}$; covering AI/ML model liability is explicitly required.

5. Security Requirements

5.1 SAST: All pipeline and API code scanned with Bandit (Python) and SonarQube. No high/critical findings at any release. Dependency scanning (SBOM) required for every release.

5.2 Annual Security Audit: Annual third-party security audit of the ML platform covering data access controls, model serving infrastructure, and API security. Audit report shared with Client within 30 days of completion.

5.3 Data Privacy: Subscriber PII must be pseudonymised before model training. All model inputs and outputs logged with 90-day retention. DPDP Act 2023 compliance required. Data processing agreement signed before any data is transferred.

6. Project Closure

Closure: (1) Model AUC ≥ 0.87 in production for 30 days. (2) Churn rate reduction $\geq 10\%$ demonstrated (15% target over 6 months). (3) Bias audit passed. (4) SAST scan clean. (5) Monitoring dashboard operational. (6) Handover documentation complete including model card, data dictionary, and retraining runbook.